🏠 Builder Basics — The Sloped Roof (Extension)

Topic: Variables · **Course:** Builder Basics · **Type:** Extension Activity · **Time:** about 45 minutes

> 🧩 This is a bonus challenge, not a main lesson. Try it once you are comfortable building shapes and you are ready to think about how one number can do a lot of work.

1 · Predict 🔮

In the **Sloped Roof** world, typing `slope` builds a stepped roof. It stacks four square **rings** of stairs, each smaller than the one below, so the roof climbs up to a point.

A variable called `roof_len` holds the side length of each ring. It **starts at 7** and **loses 2** each ring, shrinking until the roof reaches its peak.

Write the four values of `roof_len`, from the bottom ring to the top ring.

Each ring has four sides. To estimate the stairs in a ring, think of about four times the side length. Add up the four sides — 7, then 5, then 3, then 1 — and write the rough total number of stairs in the whole roof. Show your adding.

What is the LAST value of `roof_len`, the one at the very top? What kind of shape does a ring that small make?

Meet a second variable, `step` — it decides **how much the ring shrinks** each layer. Here `step` is 2, so the side drops 7, 5, 3, 1.

If `step` were 1 instead of 2, would the roof be steeper (rises fast) or gentler (rises slowly)? Would it need MORE rings or FEWER rings to reach the peak? Explain why in a sentence or two.

2 · Trace Two Things 🔢

Each layer changes two things: the **side** gets smaller and the **height** goes up by 1. Start `roof_len` at 7 and height at 1; take 2 off the side and add 1 to the height each row.

Layer	`roof_len` (ring side)	Height
1	7	1
2		
3		
4		

Which change makes the ring smaller each layer?

Which change makes the roof taller each layer?

Which ring becomes the peak — the top point of the roof — and why is it the peak?

3 · Reason About Bugs 🧠

No code to rewrite — read what goes wrong and explain it in writing.

Bug A — The shrink happens too early. The program takes 2 off `roof_len` **before** it builds the ring, not after. So the first ring uses the shrunk number, not 7.

What goes wrong with the roof? (Think about the bottom ring.) How would you fix the order of the steps?

Bug B — The variable never changes. Imagine `roof_len` stays at 7 for every layer and never loses its 2.

What shape would you get instead of a roof? Why does the roof need the variable to update each time?

4 · Design Your Own Roof 🏗️

Plan a roof of your own — in numbers and words, not in a code box.

Pick a starting `roof_len` (the widest ring) and a `step` (how much it shrinks each layer):

List the ring sizes your numbers produce, layer by layer, until you reach the peak. Just write the numbers, in order.

How many rings does it take to reach the peak?

The big idea: the whole roof is built from one variable plus one shrink rule, instead of four rings written by hand. Explain WHY that is better. If you wanted a much bigger roof, what is the smallest thing you would change?

5 · Show Your Work 📸 🎥

Open the **Sloped Roof** world and type `slope` in the chat to run the program and watch the roof build. Then build your **own** version by changing the numbers — try a different starting `roof_len` or a different `step` and see how the roof changes shape.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.
>
> The most fun part was _____.
>
> Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.